Name:	Student ID:

BIAS, VARIANCE AND GENERALISATION PROBLEM SHEET

This sheet covers the lectures on bias and variance, over-fitting and symmetry. Question 1 practises the basic ideas, Question 2 works through shrinkage in detail and Question 3 ridge regression. Each ends with a short Python part (`import numpy as np` is assumed). Question 4 is harder and optional: it shows that gradient descent regularises, and builds an invariant function by averaging over a group.

1

(a) The same machine is trained on four different data sets. At one input $\boldsymbol{x}$, where the true output is $y = 2.5$, the four machines predict 2.0, 2.4, 1.6 and 2.0. Find the mean machine's prediction, the bias $(\hat{f}_m - y)^2$ and the variance at this input, and check that they add up to the average squared error. [4 marks]

$\overline{4}$

(b) For each pair, which machine would you expect to have the higher bias, and which the higher variance: (i) a straight line and a degree-15 polynomial fitted to 20 noisy points; (ii) k-nearest neighbours with $k = 1$ and with $k = 50$? Explain briefly. [4 marks]

$\overline{4}$

(c) You have 1000 labelled examples and use 5-fold cross validation. How many models do you train, how many examples does each see, and how many examples is each validated on? Why must you not measure performance on the training data? [3 marks]

3

(d) An ensemble averages M predictors, each with variance $s^2 = 1$ at some input and pairwise correlation $\rho = 0.3$. Using $\mathrm{Var}(\bar{f}_M) = \rho s^2 + (1-\rho)s^2/M$, find the variance for $M = 10$ and as $M \to \infty$. Does averaging change the bias? [3 marks]

3

(e) In Python, take 20 equally spaced inputs on $[0,1]$ and targets $\sin(2\pi x) + 0.3\epsilon$ with $\epsilon \sim \mathcal{N}(0,1)$. For polynomials of degree 1, 3 and 9, fit 1000 independent data sets (`np.polyfit`) and estimate the bias (squared error of the mean fit, averaged over 100 test points) and the variance. [6 marks]

6

End of question 1

(a) $\frac{}{4}$	(b) $\frac{}{4}$	(c) $\frac{}{3}$	(d) $\frac{}{3}$	(e) $\frac{}{6}$	Total $\frac{}{20}$

2 We estimate the mean θ of a distribution with variance σ^2 from m independent samples, whose sample mean is $\bar{x}$. Instead of $\bar{x}$ we use the *shrinkage* estimate $\hat{\theta} = c\bar{x}$ with $0 \leq c \leq 1$.

(a) Find $\mathbb{E}\left[\hat{\theta}\right]$ and $\mathrm{Var}(\hat{\theta})$. For which c is $\hat{\theta}$ unbiased? [4 marks]

4

(b) Show that the mean squared error is $\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = (1-c)^2\theta^2 + c^2\sigma^2/m$, and identify the bias and variance terms. [4 marks]

4

(c) Find the c^* that minimises the mean squared error and the minimum error. Show that it is smaller than σ^2/m, the error of $\bar{x}$, whenever $\theta \neq 0$. [5 marks]

5

(d) Why can we not use c^* directly? What happens to c^* as $m \to \infty$, and why does that make sense? [3 marks]

$\frac{}{3}$

(e) Take $\theta = 2$, $\sigma = 4$, $m = 10$. Compute c^* and the errors for $c = 1$ and $c = c^*$. Check by simulation in Python, drawing 100000 samples of size m. [4 marks]

$\frac{}{4}$

End of question 2

(a) $\frac{}{4}$ (b) $\frac{}{4}$ (c) $\frac{}{5}$ (d) $\frac{}{3}$ (e) $\frac{}{4}$ Total $\frac{}{20}$

3 Data are generated as $\boldsymbol{y} = \mathbf{X}\boldsymbol{w}^* + \boldsymbol{\epsilon}$, where the noise terms ϵ_k are independent with mean 0 and variance σ^2. Ridge regression minimises $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \eta\|\boldsymbol{w}\|^2$.

(a) In one dimension, $y_k = w^* x_k + \epsilon_k$ for $k = 1,\ldots,m$. With $S = \sum_k x_k^2$, show that $\hat{w} = \sum_k x_k y_k/(S+\eta)$ and find its mean and variance. For $x = (1,2,3)$, $w^* = 0.5$ and $\sigma = 1$, find the bias, variance and mean squared error for $\eta = 0$ and $\eta = 4$. [5 marks]

15

(b) In general write the SVD $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\mathsf{T}$, with columns $\boldsymbol{u}_k$ and $\boldsymbol{v}_k$ and singular values s_k. Show that $\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\mathbf{I} = \mathbf{V}(\mathbf{S}^2 + \eta\mathbf{I})\mathbf{V}^\mathsf{T}$, and hence that the component of $\hat{\boldsymbol{w}}$ along $\boldsymbol{v}_k$ is $\hat{w}_k = s_k \boldsymbol{u}_k^\mathsf{T}\boldsymbol{y}/(s_k^2+\eta)$. [5 marks]

15

(c) With $w_k^* = \boldsymbol{v}_k^\mathsf{T}\boldsymbol{w}^*$, show that $\boldsymbol{u}_k^\mathsf{T}\boldsymbol{y} = s_k w_k^* + \boldsymbol{u}_k^\mathsf{T}\boldsymbol{\epsilon}$, and hence that $\mathbb{E}\big[\hat{w}_k\big] = s_k^2 w_k^*/(s_k^2+\eta)$ and $\mathrm{Var}(\hat{w}_k) = \sigma^2 s_k^2/(s_k^2+\eta)^2$. [4 marks]

$\overline{4}$

(d) Take two components with $s = (2, 0.1)$, $w^* = (1,1)$ (in the $\boldsymbol{v}$ basis) and $\sigma = 1$, so that $\mathbb{E}\big[\|\hat{\boldsymbol{w}} - \boldsymbol{w}^*\|^2\big] = \sum_k \frac{\eta^2 w_k^{*2} + \sigma^2 s_k^2}{(s_k^2+\eta)^2}$. Evaluate it for $\eta = 0$ and $\eta = 1$. Show that each term is smallest at $\eta = \sigma^2/w_k^{*2}$, and check your values by simulation in Python. [6 marks]

$\overline{6}$

End of question 3

(a) $\overline{5}$	(b) $\overline{5}$	(c) $\overline{4}$	(d) $\overline{6}$	Total $\overline{20}$

4 This question is harder and optional.

(a) We minimise $\frac{1}{2}\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2$ by gradient descent, $\boldsymbol{w} \leftarrow \boldsymbol{w} - r\mathbf{X}^\mathsf{T}(\mathbf{X}\boldsymbol{w} - \boldsymbol{y})$, starting from $\boldsymbol{w} = \mathbf{0}$, where $\mathbf{X}$ is $m \times p$ with $p > m$ and $\mathbf{X}\mathbf{X}^\mathsf{T}$ invertible. Explain why $\boldsymbol{w}$ is always of the form $\mathbf{X}^\mathsf{T}\boldsymbol{a}$. If it converges to an exact fit, show that the limit is $\boldsymbol{w} = \mathbf{X}^\mathsf{T}(\mathbf{X}\mathbf{X}^\mathsf{T})^{-1}\boldsymbol{y}$. [6 marks]

6

(b) Show that this is the exact fit with the smallest norm. (Hint: any other exact fit is $\boldsymbol{w} + \boldsymbol{z}$ with $\mathbf{X}\boldsymbol{z} = \mathbf{0}$.) Why does this matter for over-parameterised models such as neural networks? [6 marks]

6

(c) Check part (a) in Python with $m = 10$, $p = 30$, random **X** and $\boldsymbol{y}$, learning rate $r = 0.01$ and 20000 steps. [5 marks]

5

(d) The rotations of a square image by 0, 90, 180 and 270 degrees form a group C_4. For any function f of images, show that $\bar{f}(\boldsymbol{x}) = \frac{1}{4}\sum_{k=0}^{3} f(R^k \cdot \boldsymbol{x})$, where R is a rotation by 90 degrees, is unchanged when $\boldsymbol{x}$ is rotated. Check this in Python with `np.rot90` for $f(\boldsymbol{x}) = \sum_{ij} x_{ij}(4i + j)$ on a random 4×4 image. [8 marks]

8

End of question 4

(a) ___/6	(b) ___/6	(c) ___/5	(d) ___/8	Total ___/25